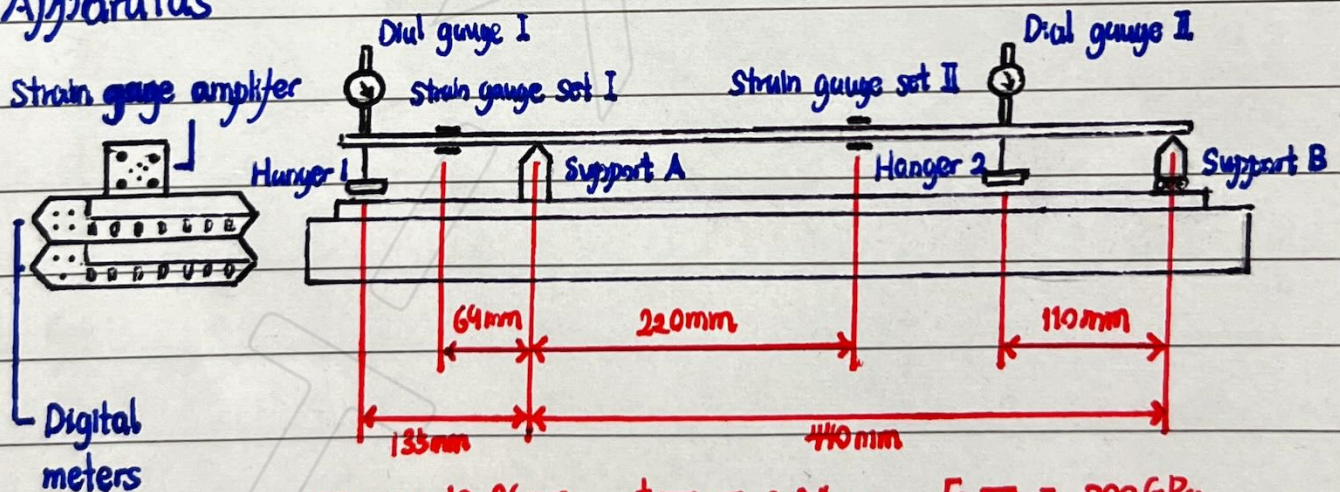


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Aiming

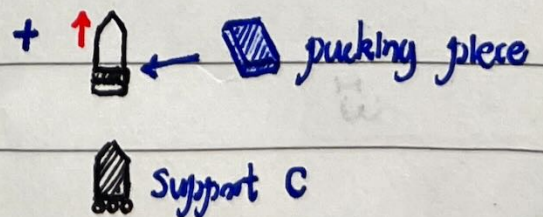
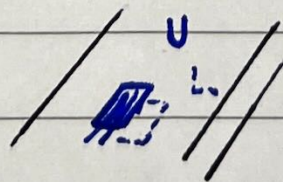
- behaviour of simple beam structures
- application of linear superposition
- response of a statically indeterminate beam
 - superposition of two statically determinate beams
- sensitivity of statically determinate/indeterminate beams
 - change the height of a support
- compare experiment with theory

Apparatus



$$W = 18.86\text{mm}, t = 2.94\text{mm}, E = 200\text{GPa}$$

* strain gauge



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Procedure

- Reset equipment
- Connect the **strain gauge**, zero the SGA output in '^{2V}~~200mV~~' scale

According to the Strain Measurement Pre-Lab

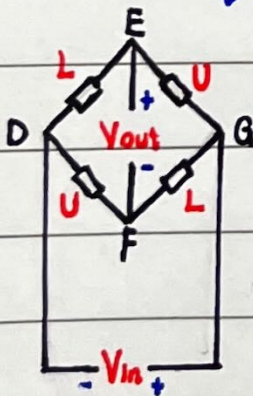
$$R = \rho \frac{L}{A}$$

If ϵ is not too large

$$\frac{\Delta R}{R} = S \epsilon$$

gauge factor = 2.12

measured by Wheatbridge since it is very small



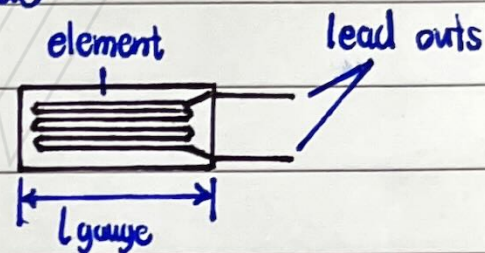
$$V_s = 1V, \text{ GAIN} = 500$$

$$V_{out} = V_{in} \left[\frac{\Delta U}{U} + \frac{\Delta L}{L} + \frac{\Delta U}{U} - \frac{\Delta L}{L} \right]$$

$$\bar{V}_{out} = \text{GAIN} \cdot V_{out}$$

$$\epsilon = \frac{\frac{\Delta R}{R}}{S} = \frac{V_{out}}{V_{in} S} = \frac{\bar{V}_{out}}{V_{in} S \cdot \text{GAIN}}$$

$$\text{If } \bar{V}_{out} = 1\text{mV}, V_{in}^{\text{out}} =$$



① Apply load to Hanger 1

→ 20N at beginning $5N \times 4$

→ remove 5N each time

→ record reading

* '~~200mV~~' scale might switched as load rises

② Apply load to Hanger 2

→ 40N at beginning $5N \times 8$

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→ remove 5N each time

→ record reading

③ Apply load to both Hangers

→ 5N to 1 → 10N to 2

→ record reading

$\times 4$

→ $5N \times 4$ at 1 → $10N \times 4$ at 2

④ Lift support A

→ insert packing piece at A

→ repeat ①

→ remove packing piece and reset support

⑤ Statically indeterminate condition

→ replace Hanger 1 by support C

→ 40N at Hanger 2 at beginning $5N \times 8$

→ remove 5N each time

→ record reading

⑥ Lift support A in statically indeterminate condition

→ repeat ④

→ repeat ⑤

Prepare an excel sheet to record the data

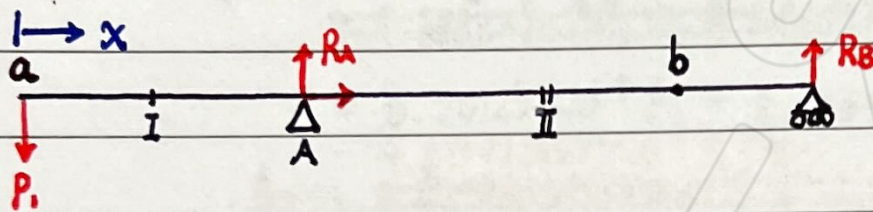
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Prediction by theory

* $\downarrow$ & $\uparrow$ positive

* calculate in m

① Apply load to Hanger 1



$$P_1 + R_A + R_B = 0$$

$$(B), 440 R_A + 575 P_1 = 0 \quad \therefore R_A = \frac{115}{88} P_1 \quad \therefore R_B = -\frac{27}{88} P_1$$

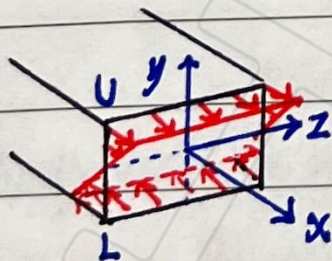
$$\therefore M = P_1 x - \frac{115}{88} P_1 \langle x - 0.135 \rangle$$

$$\therefore M_I =$$

$$M_{II} =$$

$$M_a =$$

$$M_b =$$



$$M_y = 0, M_z = M, \epsilon_x = \frac{M_z}{I_z} y = \epsilon_x E$$

$$I_z =$$

$$\therefore \epsilon = \frac{M}{I_z E} y$$

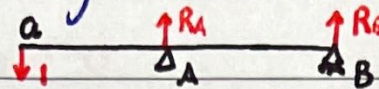
$$\therefore \epsilon_{U, I} =$$

$$\epsilon_{L, I} =$$

$$\epsilon_{U, II} =$$

$$\epsilon_{L, II} =$$

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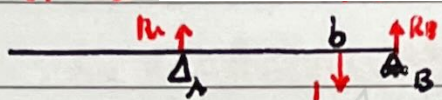
At a,  $\therefore \bar{R}_A = \frac{115}{98}, \bar{R}_B = -\frac{27}{98}$

$$\therefore \bar{M} = x - \frac{115}{98} \langle x - 0.135 \rangle$$

$$\therefore 1 \cdot \delta = \int_0^L \frac{M}{EI} \bar{M} dx$$

$$\therefore \delta EI_2 = \int_0^{0.135} P_1 x^2 dx + \int_{0.135}^{0.575} P_1 \left[x - \frac{115}{98} \langle x - 0.135 \rangle \right]^2 dx$$

$\therefore \delta a :$

At b,  $\therefore \bar{R}_A = \frac{1}{4}, \bar{R}_B = \frac{3}{4}$

$$\therefore \bar{M} = -\frac{1}{4} \langle x - 0.135 \rangle + \langle x - 0.465 \rangle$$

~~$$\delta EI_2 = \int_{0.135}^{0.465} P_1 x \left[-\frac{1}{4} \langle x - 0.135 \rangle \right] dx + \int_{0.465}^{0.575} P_1 x dx$$~~

$$\therefore \delta EI_2 =$$

$\therefore \delta b =$

② Apply load to Hanger 2

Similar to ①

$$R_A = \frac{1}{4} P_2, R_B = \frac{3}{4} P_2, M = -\frac{1}{4} P_2 \langle x - 0.135 \rangle + P_2 \langle x - 0.465 \rangle$$

$$\therefore M_1 =$$

$$M_a =$$

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$$\therefore E_{U, I} =$$

$$E_{L, I} =$$

$$E_{U, II} =$$

$$E_{L, II} =$$

$$\text{At } a, \delta EI_2 = \int_{0.135}^{0.465} P_2 \left[x - \frac{115}{28} (x - 0.135) \right] (-\frac{1}{4} (x - 0.135)) dx \\ + \int_{0.465}^{0.575} P_2 \left[(x - \frac{115}{28} (x - 0.135)) (-\frac{1}{4} (x - 0.135) + (x - 0.465)) \right] dx$$

=

=

$$\therefore \delta a =$$

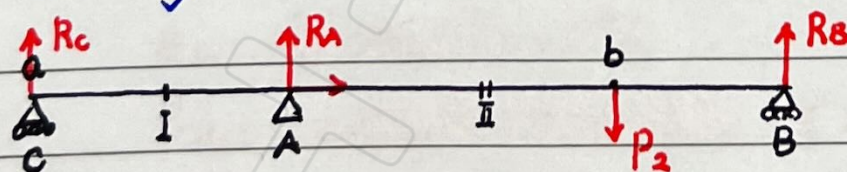
$$\text{At } b, \delta EI_2 = \int_{0.135}^{0.465} P_2 (-\frac{1}{4} (x - 0.135)^2) dx \\ + \int_{0.465}^{0.575} P_2 \left[-\frac{1}{4} (x - 0.135) + (x - 0.465) \right]^2 dx$$

=

=

$$\therefore \delta b =$$

③ statically indeterminate condition



$$\rightarrow x \quad (C), -135 R_A + 465 P_2 - 575 R_B = -27 R_A + 93 P_2 - 115 R_B = 0$$

$$(B), 110 P_2 - 440 R_A - 575 R_C = 22 P_2 - 88 R_A - 115 R_C = 0$$

$$R_C + R_A + R_B = P_2$$

$$M = EI v'' = -R_C x - R_A \langle x - 0.135 \rangle + P_2 \langle x - 0.465 \rangle$$

$$\therefore EI v' = -\frac{1}{2} R_C x^2 - \frac{1}{2} R_A \langle x - 0.135 \rangle^2 + \frac{1}{2} P_2 \langle x - 0.465 \rangle^2 + C_1$$

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$$\therefore EI v'''' = -\frac{1}{6} R_c x^3 - \frac{1}{6} R_A (x - 0.135)^3 + \frac{1}{6} P_2 (x - 0.465)^3 + C_1 x + C_2$$

When $x=0$, $v=0 \quad \therefore C_2=0$

When $x=0.575$, $v=0$

$$\therefore -515^3 R_c - 440^3 R_A + 110^3 P_2 + 3450 C_1 = 0$$

When $x=0.135$, $v=0$

$$\therefore -135^3 R_c + 810 C_1 = 0$$

$$\therefore -174630000 R_c - 85184000 R_A + 1331000 \frac{P_2}{\lambda} = 0$$

$$\therefore R_A =$$

$$R_B =$$

$$R_c =$$

$$\therefore M =$$

$$\therefore M_I =$$

$$M_{II} =$$

$$\therefore E_{U,I} =$$

$$E_{L,I} =$$

$$E_{U,II} =$$

$$E_{L,II} =$$

When $x=0.465$,

$$EI v'' =$$

$$\therefore \delta_{II(\downarrow)} =$$

3/9/2023 Simple Beams During Lab

- Correct SGA output from 200 mV to 2V scale
- Is the relationship in Wheatstone bridge valid?

$$\epsilon = \bar{V}_{out} \cdot \left(\frac{1}{1 \text{ mV}} \cdot 0.943 \mu \right)$$

* linear?

** Excited V_s is separate from the voltage that measured

- When will the approximation by theory invalid?

* Exceed the small angle approximation

- Why 6 tests carried out instead of 1?

- Any errors/uncertainties need to be considered?

- When insert support C, the beam displace according to the deflection gauge!

- Error might occurs, remember to recording

3/9/2023 Simple Beams During Lab

Uncertainty Error		Load Hanger 1 (N)	Load Hanger 2 (N)	Strain Gauge 1 (V)	Strain Gauge 2 (V)	Deflection Gauge 1 (mm)	Deflection Gauge 2 (mm)
Test 1	Loading Hanger 1						
	Unloading Hanger 1						
Test 2	Loading Hanger 2						
	Unloading Hanger 2						
Test 3	Simultaneous Loading						
	Simultaneous Unloading						
Test 4	Insert packing A and Simultaneous Loading						
	Insert packing A and Simultaneous Unloading						

↓ Hanger 1

↓ Hanger 2

↓ both hanger

↓ packing A

Statically determine condition

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Uncertainty Error		Load Hanger 1 (N)	Load Hanger 2 (N)	Strain Gauge 1 (V)	Strain Gauge 2 (V)	Deflection Gauge 1 (mm)	Deflection Gauge 2 (mm)
Test 5	Loading Hanger 2 ↓ 2						
	Unloading Hanger 2 ↓ 2						
Test 6	Insert packing A and Loading Hanger 2 ↓ 2 + Δ						
	Insert packing A and Unloading Hanger 2 ↓ 2 + Δ						

Statically indeterminate condition

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Prediction by theory

* Test 1, 2, 5 already done in Pre-Lab

* For M , $\downarrow$ positive

For ϵ , extension is positive

For δ , downward $\downarrow$ positive

① P_1 at Hanger 1

$$M = P_1 (x - \frac{115}{88} P_1 (x - 0.135))$$

$$M_I =$$

$$M_{II} =$$

$$\epsilon_{U, I}$$

$$\epsilon_{U, II}$$

$$\delta_a =$$

$$\delta_b =$$

② P_2 at Hanger 2

$$M = P_2 (-\frac{1}{4} (x - 135) + (x - 465))$$

$$M_I =$$

$$M_{II} =$$

$$\epsilon_{U, I}$$

$$\epsilon_{U, II}$$

$$\delta_a =$$

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$$\delta_b = 1.249703222 \times 10^{-1} P_2$$

③ P_1 at Hanger 1, P_2 at Hanger 2

$$E_{U,1} =$$

$$E_{U,1} =$$

$$\delta_a =$$

$$\delta_b =$$

④ Lift support A then ③

$$E_{U,1} =$$

$$E_{U,1} =$$

$$\delta_a =$$

$$\delta_b =$$

⑤ Statically indeterminate

$$M = \frac{121}{144} P_2 x - \frac{91}{144} \langle x - 135 \rangle P_2 + \langle x - 465 \rangle P_2$$

$$M_z =$$

$$M_z =$$

$$E_{U,1} =$$

$$E_{U,1} =$$

$$\delta_b =$$

⑥ Lift support A then ⑤

E changes

δ changes

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		Load Hanger 1 (N)	Load Hanger 2 (N)	Experiment Data						Prediction by theory					
				Strain 1	Absolute error	Strain 2	Absolute error	Deflection a (mm)	Absolute error	Deflection b (mm)	Absolute error	Strain 1	Strain 2	Deflection a (mm)	Deflection b (mm)
Test 1	Loading Hanger 1														
	Unloading Hanger 1														
Test 2	Loading Hanger 2														
	Unloading Hanger 2														
Test 3	Simultaneous Loading														
	Simultaneous Unloading														
Test 4	Insert packing A and Simultaneous Loading														
	Insert packing A and Simultaneous Unloading														

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dp

dp

I3

I3

Test 1

Test 2

Test 3

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d_1

d_1

ϵ_1

ϵ_1

Test 3

Test 4

5

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		Experiment Data						Prediction by theory		
		Load Hanger 1 (N)	Load Hanger 2 (N)	Strain 1	Absolute error	Strain 2	Absolute error	Deflection a (mm)	Absolut error	Deflection b (mm)
Test 5	Loading Hanger 2									
	Unloading Hanger 2									
Test 6	Insert packing A and Loading Hanger 2									
	Insert packing A and Unloading Hanger 2									

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d_2

d_1

ϵ_2

ϵ_1

Test 5

Test 6

x7

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Validation Plot matches theory from experiments ✓

Effect of Raising Support see previous page (page 7 Pre-lab)

statically determinate strain don't change, displacement change

statically indeterminate strain and displacement change

Validation of superposition

mark for statically determinate beam (Test 3 = Test 1 + Test 2)

Error analysis

- see during-lab parts
- some possible source of error might occurs
 - disturbances on the ^{工作台} workbench as people lean/hit it
 - uneven loading and sudden jolts to the equipment when loading and unloading the weights may lead errors
 - ★ small angle approximation becomes invalid at higher loads
 - ★ hysteresis 迟滞现象
(which can be observed from the data)
 - imperfections in the cross-section of the beam from material cause the different E across the beam

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Further improvement

- If can, consider if the incremental merit justifying the effort
i.e. consider the error from measuring of the weights

Addition points

About hysteresis:

- a. if there are any valid sources of hysteresis:
 1. beam strain a lot;
 2. repeating loading leads fatigue;
 3. loading and unloading;
- b. at what relative proportions? that their effects are.